

Technical University of Denmark
Department of Applied Mathematics and Computer Science

02170 Database Systems
Practice Examination Part 1 — Exam C

Aids allowed: All
Duration: 3 hours
Weighting: Part 1: 50%, Part 2: 50%

Note: unless explicitly asked for, SQL queries should not return tables having duplicate rows.

A database for a Music Platform

A database named **MusicDB** contains information about record labels, artists, songs, and playlists.

A **Label** has a unique name (*lname*) and is based in a country (*country*). An **Artist** has a unique id (*aid*), a name (*aname*), and is signed to a single label (identified by *lname*). A **Song** has a unique id (*songid*), a title (*stitle*), and a release year (*syear*).

The **performs** relation records which artists perform which songs. A song can be performed by multiple artists (collaboration). The **Playlist** relation represents curated playlists, each owned by a label. The **includes** relation records which songs are included in which playlists.

Relation schemas:

Label(lname, country)

Song(songid, stitle, syear)

Playlist(pname, lname)

Artist(aid, aname, lname)

performs(aid, songid)

includes(pname, songid)

Label

lname	country
NordicSound	Denmark
CopenhagenBeat	Denmark
StockholmRec	Sweden
GlobalTunes	USA

Artist

aid	aname	lname
1	MØ	NordicSound
2	Lukas G.	NordicSound
3	Robyn	StockholmRec
4	Nephew	CopenhagenBeat
5	Billie E.	GlobalTunes

Song

songid	stitle	syear
S1	Final Song	2021
S2	Dancing On My Own	2010
S3	Lean On	2015
S4	Dark Horse	2022

performs

aid	songid
1	S1
2	S1
2	S3
3	S2
4	S4
1	S3

Playlist

pname	lname
DanishHits	NordicSound
NordicVibes	CopenhagenBeat
GlobalTop	GlobalTunes

includes

pname	songid
DanishHits	S1
DanishHits	S3
NordicVibes	S4
GlobalTop	S1
GlobalTop	S2

Questions

Question 1

Write an SQL query that returns a table containing, for each artist who is signed to a **Danish** label and who has **never performed any song**, his/her **aname** and **lname**.

Question 2

Write an SQL query that returns a table containing, for each song performed by **2 or more artists**, the **songid**, **stitle**, **syear**, and the total number of artists who perform it. Name the count **numArtists**.

Question 3

Write an SQL query that returns a table containing, for each playlist, the **pname** and the number of **distinct songs performed by at least one artist signed to the same label as the playlist**. Name the count **numSongs**.

Important: every playlist must appear in the result, even those with no matching songs.

Question 4

Create an SQL function named **songCount**, which takes the **aid** (of type **INT**) of an artist as argument and returns the number of songs performed by that artist. If the artist does not exist, it should return 0. The returned type should be **INT**.

Do not include any DROP FUNCTION or SELECT statements using the function.

Answers Template (answers_C.sql)

```
-- Student ID:  
USE MusicDB;  
  
-- q1:  
-- q2:  
-- q3:  
-- q4:
```